EdgeDAM: Bounding-Box Distractor-Aware Memory for Real-Time Object Tracking

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Abstract

Robust single-object tracking often fails when occlusion, fast motion, or visually similar distractors interrupt the target evidence. Memory-augmented trackers address this problem by storing past target states for later recovery, but recent distractor-aware designs largely rely on dense segmentation features, mask propagation, and attention-based retrieval, making them computationally expensive for real-time bounding-box tracking. We propose **E**dge**D**AM, a detection-guided tracking framework that reformulates distractor-aware memory at the bounding-box level. Instead of maintaining dense mask tokens, EdgeDAM introduces a compact dual-buffer memory: a Recent-Aware Memory stores geometrically verified target states, while a Distractor-Resolving Memory preserves stable appearance anchors and penalizes hard negative candidates during re-identification. Reliability-driven switching and held-box stabilization invoke recovery only under uncertain tracking states, preventing distractor contamination during occlusion. EdgeDAM requires no segmentation masks, transformer memory attention, or dense feature retrieval. Across six benchmarks, it achieves 0.741 quality, 0.730 IoU, and 0.973 robustness on DiDi, while running at 25 FPS on mobile device.

1 Introduction

Visual object tracking (VOT) addresses the problem of localizing a target initialized in the first frame. Existing trackers are predominantly developed for RGB-based visual streams and are widely used in autonomous driving, drone imaging, intelligent surveillance, and other real-world applications [3, 10, 12]. Despite this progress, robust tracking remains difficult in complex scenes where fast motion, illumination variation, background clutter, visually similar distractors, and out-of-view events interrupt reliable target evidence. These challenges are not uniformly distributed across a sequence; instead, they occur in short and critical intervals that can trigger identity switches and long-term drift. Therefore, improving tracking robustness cannot be achieved only by adding more labeled data, but requires mechanisms that can detect unreliable tracking states, preserve trustworthy target evidence, and recover the target when distractors or occlusion corrupt the current observation.

Despite substantial progress in modern tracking frameworks [18, 19, 22], robust single-object tracking remains difficult when the visual evidence of the target becomes unreliable. Occlusion, fast motion, deformation, and visually similar distractors can weaken the target response and introduce competing hypotheses. Once a tracker selects a distractor instead of the true target, this error can propagate through subsequent updates, leading to long-term identity drift.

Memory-augmented trackers address this problem by storing past target states and retrieving them when current-frame evidence becomes ambiguous. This design is especially effective under occlusion and distractor interference, where the tracker must recover the target after it is partially or fully hidden. However, recent distractor-aware memory mechanisms are largely inherited from segmentation-based tracking. Methods such as SAM2.1++ [18] and SAMURAI [19] maintain dense mask features, propagate object masks across frames, and retrieve memory through attention over high-dimensional visual tokens. While powerful, these operations introduce substantial computational overhead and are not naturally matched to the output space of bounding-box tracking. As shown in Fig. 1, such memory-heavy trackers operate at only 2–8 FPS even on high-end GPUs, making them unsuitable for real-time mobile deployment.

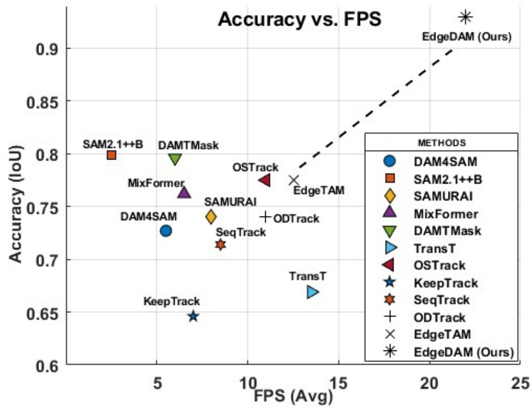

Figure 1: Accuracy–efficiency comparison of representative trackers. Robust memory-based designs improve occlusion handling but incur high computational cost, while lightweight trackers sustain real-time throughput yet remain vulnerable to distractors.

At the other extreme, efficient bounding-box trackers such as EdgeTAM [22] and OSTrack [20] achieve higher throughput but remain fragile when target evidence is interrupted. These trackers avoid dense memory and mask propagation, but they typically lack an explicit mechanism for separating reliable target states from visually similar distractors. As a result, they can maintain speed but often sacrifice robustness under occlusion-heavy or distractor-rich conditions. This exposes a deeper limitation in current VOT research: the robustness efficiency gap is not only a matter of computational cost, but also a mismatch between memory representation and tracking objective. Existing distractor-aware memory is designed around dense segmentation states, whereas bounding-box tracking only requires recovery of a compact target state.

This observation motivates a different formulation. For bounding-box tracking, distractor-aware reasoning does not necessarily require dense masks, transformer memory attention, or global feature retrieval. Since the prediction target is a low-dimensional box state, memory can instead be represented as a compact set of reliability-verified target hypotheses and explicitly separated hard-negative distractor states. In this view, the key problem is not how to compress a segmentation memory bank, but how to redesign memory itself for the bounding-box tracking space. This leads to the central question of this work: can distractor-aware memory be reformulated at the bounding-box level while preserving robustness under occlusion and distractors?

We answer this question with **EdgeDAM**, a detection-guided single-object tracking framework that reformulates distractor-aware memory from dense segmentation-token storage into compact bounding-box state memory. Instead of maintaining dense mask tokens, EdgeDAM introduces a dual-buffer *Distractor-Aware Memory (DAM)* that explicitly separates positive target evidence from hard negative distractor evidence. The **Recent-Aware Memory (RAM)** stores geometrically verified target states using box-level consistency criteria, while the **Distractor-Resolving Memory (DRM)** maintains hard-negative candidates and penalizes their re-selection during re-identification. This positive–negative memory factorization allows EdgeDAM to perform distractor-aware recovery without segmentation masks, mask propagation, transformer memory attention, or dense feature retrieval.

EdgeDAM further introduces a contamination-aware recovery strategy for uncertain tracking states. Rather than invoking detection and memory retrieval at every frame, EdgeDAM uses confidence-driven switching to activate recovery only when the tracker becomes unreliable. During occlusion, a held-box stabilization mechanism temporarily preserves and expands the last reliable estimate, preventing ambiguous observations from corrupting memory. When the target reappears, RAM provides temporally consistent target hypotheses, while DRM suppresses distractor candidates that are visually or geometrically likely to cause identity switches. Thus, EdgeDAM treats memory not as a dense feature store, but as a compact box-level decision mechanism for deciding when to trust, hold, recover, or reject candidate states.

We instantiate this formulation in a lightweight detector-guided tracking pipeline. The detector is trained as a category-agnostic single-class object detector by merging all object categories into one target index, enabling the same tracking framework to operate across standard benchmarks without category-specific modification. In our implementation, this design enables real-time tracking on mobile hardware while preserving robustness under occlusion and distractor interference.

Our main contributions are:

- We reformulate distractor-aware memory for single-object tracking from dense segmentation token storage to compact bounding-box state memory, showing that robust distractor reasoning can be achieved without mask propagation or transformer memory attention.

- We introduce a dual-factor memory design that explicitly separates positive target evidence from hard negative distractor evidence. The Recent-Aware Memory admits only reliability-verified target states, while the Distractor-Resolving Memory suppresses distractor re-selection during re-identification.

- We propose a contamination-aware recovery mechanism based on confidence-driven switching and held-box stabilization, which prevents unreliable observations from corrupting memory during occlusion and fast motion.

- We instantiate the formulation in an edge-deployable detection-guided tracker and demonstrate strong robustness across benchmarks while maintaining real-time performance on mobile hardware.

2 Related Work

Visual object tracking has been studied through Siamese matching, regression-based localization, and transformer-based modeling. Siamese trackers [2, 21] localize the target by matching a template against a search region, while regression-based methods [7] directly estimate the target state. Transformer trackers [4, 5, 6, 15] improve target association through attention-based feature interaction. Although effective, these approaches can still drift when target evidence is corrupted by occlusion or visually similar distractors, and their high-dimensional feature matching often limits real-time deployment.

Memory-based tracking addresses such failures by storing past target evidence for recovery. Recent segmentation-oriented trackers, including SAM2 [17], SAMURAI [19], EdgeTAM [22], and SAM2.1++ [18], maintain mask features or memory tokens and retrieve them through dense attention or global memory fusion. SAM2 propagates object masks with memory attention, EdgeTAM compresses spatial memory for on-device segmentation, and SAM2.1++ introduces distractor-aware memory for robust mask tracking. However, these methods inherit a dense segmentation representation: memory is built around masks, high-dimensional tokens, and attention-based retrieval. This is effective for mask propagation but costly for real-time bounding-box tracking, where the prediction target is a compact box state.

Efficient trackers and detection-driven methods reduce this overhead by operating directly on bounding boxes. Methods such as KeepTrack [16], TransT [4], and MixFormer [5] improve target association and throughput, but they do not explicitly reformulate distractor-aware memory for the box-level tracking space. As a result, they remain vulnerable when multiple candidate boxes compete under occlusion or distractor overlap.

EdgeDAM differs from both directions. Instead of compressing dense segmentation memory or simply combining detection with short-term tracking, it reformulates distractor-aware memory as compact box-state reasoning. RAM stores reliability-verified target states, while DRM maintains hard negative distractor evidence to penalize distractor re-selection during re-identification. This target–distractor factorization enables memory-guided recovery without segmentation masks, dense feature retrieval, or transformer memory attention.

3 EdgeDAM

Given a tracking sequence $\{I_t\}_{t=1}^{T}$ and target initialization b_1 at $t = 1$, EdgeDAM estimates the target state b_t for each $t > 1$ while suppressing occlusion-induced drift and distractor-driven identity switches. The key idea is a box-level Distractor-Aware Memory (DAM) that reformulates distractor-aware tracking from dense segmentation-token memory into compact target–distractor state factorization as shown in Fig.2. Instead of storing masks, dense visual tokens, or transformer memory states, EdgeDAM maintains reliability-verified target hypotheses and hard negative distractor evidence directly in the bounding-box state space.

EdgeDAM instantiates this formulation with detector-generated hypotheses, short-term state propagation, and DAM-based state admission. The detector $\mathcal{G}$ proposes candidate boxes, the tracker $\mathcal{T}$ propagates the previous target state, and DAM determines whether the propagated hypothesis is admitted, held, or replaced by a recovered candidate. Distractor-aware reasoning is invoked only when reliability verification fails, avoiding dense memory retrieval during stable tracking.

3.1 Overview

At time step t, the detector $\mathcal{G}$ produces candidate boxes with confidence scores:

$$\mathcal{B}_t = \{(b_i^t, s_i^t)\}_{i=1}^{N_t}, \qquad s_i^t \in [0,1]. \tag{1}$$

Candidates below the detection threshold are removed:

$$\mathcal{B}_t^+ = \{(b_i^t, s_i^t) \in \mathcal{B}_t \mid s_i^t \geq \tau_s\}. \tag{2}$$

The target initialization b_1 initializes $\mathcal{T}$, the appearance template, optical-flow points, and the DAM memory. At time step t, $\mathcal{T}$ predicts a propagated hypothesis $\hat{b}_t^{\text{trk}}$. EdgeDAM

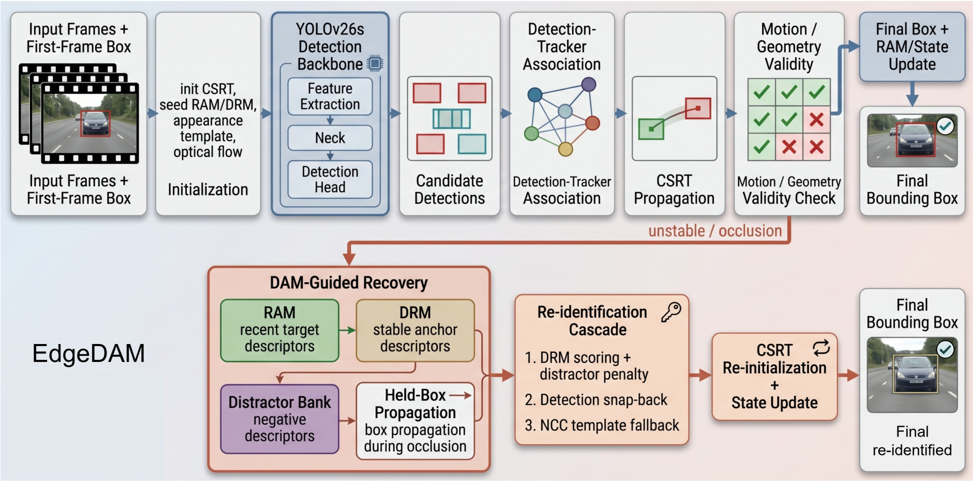

Figure 2: EdgeDAM framework overview. A YOLOv26s detector generates candidate boxes, while CSRT propagates the target state. Reliable states are admitted into RAM and used for state update. Under occlusion or instability, EdgeDAM activates DAM-guided recovery, where RAM/DRM target evidence and distractor-bank penalties score current candidates. Held-box stabilization preserves the last reliable target estimate during ambiguous states and prevents uncertain observations from contaminating memory.

evaluates $\hat{b}_t^{\text{trk}}$ using motion agreement, geometric continuity, area consistency, and candidate ambiguity. If the hypothesis satisfies reliability verification, it is accepted as $\hat{b}_t$ and admitted into RAM. Otherwise, state admission is blocked and EdgeDAM activates contamination-aware holding or DAM-guided re-identification.

Short-term propagation preserves local continuity; DAM enforces reliability-constrained memory admission. This prevents drifted or distractor-contaminated hypotheses from becoming future target evidence. Fig. 2 summarizes the overall EdgeDAM workflow.

3.2 Box-Level Distractor-Aware Memory

Let $c(b)$ and $a(b)$ denote the center and area of box b. EdgeDAM maintains a compact DAM state:

$$\mathcal{M}_t^{\text{DAM}} = (\mathcal{R}_t, \mathcal{D}_t)\,, \tag{3}$$

where $\mathcal{R}_t$ is the *Recent-Aware Memory (RAM)* and $\mathcal{D}_t$ is the *Distractor-Resolving Memory (DRM)*. RAM stores recently verified target states. DRM decomposes into stable target anchors and hard negative distractor evidence:

$$\mathcal{D}_t = (\mathcal{A}_t, \mathcal{N}_t)\,. \tag{4}$$

Here, $\mathcal{A}_t = \{(a_k, \psi_k, \rho_k)\}_{k=1}^{|\mathcal{A}_t|}$ contains stable target anchors promoted from RAM, where a_k is the anchor box, ψ_k is its descriptor, and ρ_k is its insertion time. The negative memory $\mathcal{N}_t = \{v_\ell\}_{\ell=1}^{|\mathcal{N}_t|}$ stores descriptors of ambiguous or rejected distractor candidates. DAM therefore factorizes memory into positive target evidence and negative distractor evidence, enabling candidate-level re-identification without dense mask memory.

For a box hypothesis b, EdgeDAM extracts a box-conditioned descriptor:

$$\phi(I_t,b) = \frac{[\mathbf{p}(I_t,b),\mathbf{h}(I_t,b)]}{\|[\mathbf{p}(I_t,b),\mathbf{h}(I_t,b)]\|_2+\varepsilon}, \tag{5}$$

where $\mathbf{p}$ is a resized grayscale crop and $\mathbf{h}$ is an HSV histogram. The descriptor provides compact appearance evidence; candidate acceptance is determined jointly by appearance consistency, motion agreement, box geometry, and distractor suppression.

RAM is defined as:

$$\mathcal{R}_t = \{(r_j,\phi_j,q_j,\eta_j)\}_{j=1}^{|\mathcal{R}_t|}, \tag{6}$$

where r_j is a verified target state, ϕ_j is its descriptor, q_j is its admission confidence, and η_j is its time index. At initialization,

$$\mathcal{R}_1 = \{(b_1,\phi(I_1,b_1),1,1)\}, \quad \mathcal{A}_1 = \{(b_1,\phi(I_1,b_1),1)\}, \quad \mathcal{N}_1 = \emptyset. \tag{7}$$

A state enters RAM only after satisfying the reliability constraints in Sec. 3.3; RAM is therefore a filtered target-state memory rather than an unconditional cache.

The running target area is estimated from verified RAM states:

$$\bar{a}_t = \begin{cases} \frac{1}{|\mathcal{R}_t|}\sum_{(r_j,\phi_j,q_j,\eta_j)\in\mathcal{R}_t} a(r_j), & |\mathcal{R}_t|>0, \\ a(\hat{b}_{t-1}), & |\mathcal{R}_t|=0. \end{cases} \tag{8}$$

Stable anchors are promoted from RAM into $\mathcal{A}_t$ only under local descriptor consistency:

$$\sum_{j\in\mathcal{W}_t} \mathbb{I}[\cos(\phi_j,\phi_t)\geq\tau_{\text{sim}}]\geq m_{\min}, \tag{9}$$

where $\mathcal{W}_t$ is a recent temporal window and $\phi_t=\phi(I_t,\hat{b}_t)$. This criterion rejects transient drift, partial occlusion artifacts, and distractor-contaminated states before they enter long-term target memory.

The negative memory $\mathcal{N}_t$ stores hard distractor descriptors. Target-neighborhood ambiguity is defined as:

$$\Omega(b_i^t,\hat{b}_{t-1}) = \mathbb{I}\left[\mathrm{IoU}(b_i^t,\hat{b}_{t-1})\geq\tau_{\text{occ}} \vee \frac{\|c(b_i^t)-c(\hat{b}_{t-1})\|_2}{\sqrt{a(\hat{b}_{t-1})+\varepsilon}}\leq\tau_c\right]. \tag{10}$$

Candidate reliability is defined as:

$$\Gamma(b_i^t) = \mathbb{I}\left[s_i^t\geq\tau_s \wedge \mathrm{IoU}(b_i^t,\hat{b}_{t-1})\geq\tau_i \wedge \frac{|a(b_i^t)-\bar{a}_t|}{\bar{a}_t+\varepsilon}\leq\tau_a\right]. \tag{11}$$

A candidate is inserted into $\mathcal{N}_t$ when it lies near the target hypothesis but fails the reliability test:

$$\mathcal{N}_t \leftarrow \mathcal{N}_t \cup \left\{\phi(I_t,b_i^t) \mid b_i^t\in\mathcal{B}_t^+, \Omega(b_i^t,\hat{b}_{t-1})=1, \Gamma(b_i^t)=0\right\}. \tag{12}$$

Descriptors in $\mathcal{N}_t$ are never used as target templates; they only penalize distractor-like hypotheses during recovery.

3.3 Reliability-Constrained State Validation

EdgeDAM verifies each propagated hypothesis before memory update. Let $q_t^{\mathrm{trk}} \in \{0,1\}$ denote tracker validity:

$$q_t^{\mathrm{trk}} = \mathbb{I}\left[a(\hat{b}_t^{\mathrm{trk}}) > 0 \wedge \mathrm{finite}(\hat{b}_t^{\mathrm{trk}}) \wedge \frac{\|c(\hat{b}_t^{\mathrm{trk}}) - c(\hat{b}_{t-1})\|_2}{\sqrt{a(\hat{b}_{t-1})} + \varepsilon} \le \tau_{\mathrm{jump}} \right]. \tag{13}$$

where $\mathrm{finite}(\cdot)$ verifies that all box coordinates are valid. Sparse Lucas–Kanade optical flow estimates the short-term displacement v_t, giving the motion-predicted center:

$$c_t^{\mathrm{flow}} = c(\hat{b}_{t-1}) + v_t. \tag{14}$$

The propagation-flow disagreement is:

$$d_t^{\mathrm{flow}} = \left\| c(\hat{b}_t^{\mathrm{trk}}) - c_t^{\mathrm{flow}} \right\|_2. \tag{15}$$

Candidate ambiguity is measured by the overlap set around the previous accepted state:

$$\mathcal{O}_t = \left\{ b_i^t \mid (b_i^t, s_i^t) \in \mathcal{B}_t^+, \mathrm{IoU}(b_i^t, \hat{b}_{t-1}) \ge \tau_{\mathrm{occ}} \right\}. \tag{16}$$

The propagated hypothesis is rejected when:

$$\delta_t^{\mathrm{sw}} = \mathbb{I}\left[q_t^{\mathrm{trk}} = 0 \vee d_t^{\mathrm{flow}} > \tau_f \vee \mathrm{IoU}(\hat{b}_t^{\mathrm{trk}}, \hat{b}_{t-1}) < \tau_i \vee \Delta a_t > \tau_a \vee |\mathcal{O}_t| \ge 2 \right], \tag{17}$$

with relative area variation:

$$\Delta a_t = \frac{|a(\hat{b}_t^{\mathrm{trk}}) - a(\hat{b}_{t-1})|}{a(\hat{b}_{t-1}) + \varepsilon}. \tag{18}$$

If $\delta_t^{\mathrm{sw}} = 0$, EdgeDAM accepts $\hat{b}_t^{\mathrm{trk}}$ as $\hat{b}_t$ and admits it into RAM. If $\delta_t^{\mathrm{sw}} = 1$, EdgeDAM blocks memory update and enters holding or re-identification. This state-admission rule preserves memory integrity by preventing unreliable hypotheses from contaminating target memory.

3.4 Contamination-Aware Holding

When the propagated hypothesis is unreliable, snapping to the nearest candidate can induce an identity switch. EdgeDAM instead maintains a held state b_t^{hold} anchored to the last reliable target estimate. The held-state center follows recent motion:

$$c_t^{\mathrm{hold}} = c(b_{t-1}^{\mathrm{hold}}) + \lambda_v v_{t-1}. \tag{19}$$

Its spatial extent is expanded toward the envelope of ambiguous candidates:

$$(\bar{w}_t, \bar{h}_t) = \begin{cases} \frac{1}{|\mathcal{O}_t|} \sum_{b_i^t \in \mathcal{O}_t} (w_i^t, h_i^t), & |\mathcal{O}_t| > 0, \\ (w_{t-1}, h_{t-1}), & |\mathcal{O}_t| = 0, \end{cases} \tag{20}$$

$$(w_t^{\mathrm{hold}}, h_t^{\mathrm{hold}}) = (1-\beta)(w_{t-1}, h_{t-1}) + \beta(\bar{w}_t, \bar{h}_t). \tag{21}$$

The held state is excluded from RAM and DRM. It only defines the recovery reference:

$$b_t^{\mathrm{ref}} = \begin{cases} b_t^{\mathrm{hold}}, & \text{if holding is active,} \\ \hat{b}_{t-1}, & \text{otherwise.} \end{cases} \tag{22}$$

Thus, ambiguous target evidence can guide recovery without being promoted into verified memory.

3.5 DAM-Guided Re-identification

Recovery is performed over current candidate hypotheses rather than stored memory anchors. Since anchors encode past reliable target evidence, they serve as references for scoring candidates, not as recovered states. We use the convention that a maximum over an empty memory is zero. For each candidate $b_i^t \in \mathcal{B}_t^+$, EdgeDAM computes a positive consistency score:

$$S^+(b_i^t) = \lambda_{\text{ram}} \max_{r_j \in \mathcal{R}_t} \cos(\phi(I_t, b_i^t), \phi_j) + \lambda_{\text{anc}} \max_{a_k \in \mathcal{A}_t} \cos(\phi(I_t, b_i^t), \psi_k) + \lambda_{\text{iou}} \text{IoU}(b_i^t, b_t^{\text{ref}}) + \lambda_{\text{area}} \exp\left(-\left|\log \frac{a(b_i^t)}{\bar{a}_t + \varepsilon}\right|\right). \tag{23}$$

The RAM and anchor terms measure appearance consistency with recent and stable target evidence, while the IoU and area terms enforce spatial and scale compatibility with the recovery reference.

Distractor evidence imposes a negative consistency penalty:

$$S^-(b_i^t) = \gamma \max_{\nu_\ell \in \mathcal{N}_t} \cos(\phi(I_t, b_i^t), \nu_\ell). \tag{24}$$

The final recovery score is:

$$\tilde{S}(b_i^t) = S^+(b_i^t) - S^-(b_i^t). \tag{25}$$

The recovered state is:

$$b_t^\star = \arg\max_{b_i^t \in \mathcal{B}_t^+} \tilde{S}(b_i^t). \tag{26}$$

The candidate is accepted only if it satisfies an absolute recovery threshold and a margin over the strongest competing hypothesis:

$$\tilde{S}(b_t^\star) \geq \tau_{\text{rec}}, \qquad \tilde{S}(b_t^\star) - \max_{b_i^t \neq b_t^\star} \tilde{S}(b_i^t) \geq m_{\text{rec}}. \tag{27}$$

If Eq. 27 holds, $b_t^{\text{acc}} = b_t^\star$ and $\mathcal{T}$ is re-initialized from b_t^{acc}. If DAM-guided recovery is inconclusive, EdgeDAM performs detector-guided re-association:

$$S^{\text{det}}(b_i^t) = \mu_{\text{app}} \max_{r_j \in \mathcal{R}_t} \cos(\phi(I_t, b_i^t), \phi_j) + \mu_{\text{iou}} \text{IoU}(b_i^t, b_t^{\text{ref}}) + \mu_{\text{area}} \exp\left(-\left|\log \frac{a(b_i^t)}{\bar{a}_t + \varepsilon}\right|\right), \tag{28}$$

and accepts the best candidate if $\max_{b_i^t \in \mathcal{B}_t^+} S^{\text{det}}(b_i^t) \geq \tau_{\text{det}}$. If $\mathcal{B}_t^+ = \emptyset$ or no candidate satisfies the recovery criteria, EdgeDAM applies local template-consistency search within an expanded region around b_t^{ref} and accepts the best match only if its normalized cross-correlation exceeds τ_{ncc}.

3.6 State Update

Let b_t^{acc} denote the accepted state from stable propagation, detector association, DAM-guided re-identification, detector-guided re-association, or local template-consistency search. EdgeDAM updates:

$$\hat{b}_t = b_t^{\text{acc}}, \qquad \mathcal{T} \leftarrow \text{Reinit}(\mathcal{T}, I_t, \hat{b}_t). \tag{29}$$

The accepted state is admitted into RAM, the appearance template is refreshed, and optical-flow points are re-initialized. If recent RAM descriptors satisfy Eq. 9, the latest verified state is promoted into the stable anchor set $\mathcal{A}_t$. Ambiguous candidates rejected during recovery are inserted into $\mathcal{N}_t$ and later penalize distractor-like hypotheses.

If no candidate satisfies the recovery criteria, EdgeDAM retains b_t^{hold} for at most K_{rec} recovery steps. During this interval, RAM and DRM are not updated. If no valid state is recovered after K_{rec} steps, EdgeDAM reports the held state and resumes recovery at the next step without memory admission.

4 Experiments

4.1 Implementation Details

4.1.1 Training Protocol

For benchmark evaluation, we train a single YOLOv26s detector from scratch on the union of the permitted training splits of the source tracking datasets and use it for all public benchmark evaluations without benchmark-specific fine-tuning. All foreground categories are collapsed into one target label after the official train/test splits are fixed, preserving the category-separation protocols of GOT-10k and LaSOText. Mask and polygon annotations are converted to axis-aligned boxes, and validation/test annotations are excluded from training.

EdgeDAM is evaluated with the official or publicly released toolkit for each benchmark. Baselines with public code/checkpoints are rerun under the same metric definitions; otherwise, we report results from the corresponding papers or official benchmark reports. Hyperparameters are selected once on validation data and fixed for all test evaluations. For mobile testing only, we train a separate single-class horse detector from COCO horse samples and custom horse-rider data; this model is not used for public benchmark results.

For each benchmark, training uses only its permitted training split, and evaluation follows the corresponding benchmark protocol and metric definition. The tracking pipeline is detector-agnostic: the DAM, switching, and recovery modules operate on detector boxes and can be paired with any YOLO backbone.

4.1.2 Inference Configuration

All benchmark experiments use 640×640 detector inputs, confidence threshold $\tau_s = 0.45$, NMS threshold 0.50, detection stride $\Delta = 3$, and ROI crop scale $\kappa = 2.0$. Model fusion and half-precision inference are enabled for detector execution. DAM uses $|\mathcal{R}_t| = 10$ and $|\mathcal{D}_t| = 10$ with FIFO replacement. Each memory entry stores a 16×16 grayscale crop concatenated with a 16×16 HSV histogram, followed by ℓ_2 normalization.

The reliability switch uses confidence and state-displacement thresholds $\tau_{\text{conf}} = 0.60$ and $\tau_{\text{jump}} = 0.30$. RAM admission uses $\tau_{\text{in}} = 0.50$ and area tolerance $\tau_a = 0.20$. Candidate ambiguity is detected with $\tau_{\text{occ}} = 0.40$. DRM promotion requires descriptor consistency of $\tau_{\text{sim}} = 0.85$ over a window of $W = 5$ states with $m_{\min} = 3$. The distractor penalty is fixed to $\gamma = 0.25$. After validation-based selection, all thresholds are kept fixed and are not tuned on benchmark test splits.

4.1.3 Mobile Deployment

For on-device deployment, the YOLOv26s detector is exported to ONNX and compiled to CoreML on an iPhone 15 Pro Max. The reported speed is measured end-to-end and includes detector inference, CSRT propagation, optical-flow update, candidate association, RAM/-DRM scoring, memory update, and held-box recovery. EdgeDAM uses fixed 640×640

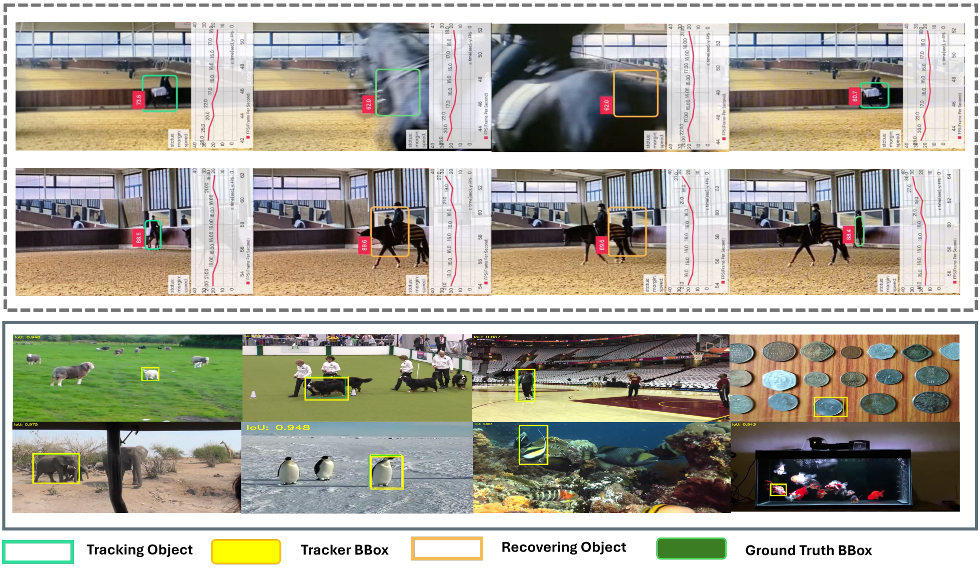

Figure 3: Top two rows (dashed border) show real-time tracking performance of EdgeDAM under occlusion on a mobile device (iPhone 15). Bottom two rows (solid border) present EdgeDAM performance on standard SOTA benchmarks.

inputs, half-precision CoreML execution, fused detector layers, adaptive detection stride, and ROI-crop inference during stable tracking. Under this complete pipeline, EdgeDAM runs at 25 FPS on-device.

4.2 Datasets

We evaluate EdgeDAM on six tracking benchmarks that stress complementary failure modes. DiDi [18] emphasizes distractor-heavy target recovery; VOT2020 [13] and VOT2022 [14] measure short-term tracking under occlusion, abrupt motion, and distractor interference; LaSOT [8] evaluates long-horizon tracking; LaSOText [9] extends evaluation to unseen categories; and GOT-10k [11] tests category-level generalization under its official zero-overlap protocol.

All benchmarks are evaluated in the bounding-box setting. We report the official or commonly adopted metrics for each benchmark: Quality, IoU, and Robustness on DiDi; EAO, Accuracy/IoU, and Robustness on VOT2020/2022; AUC on LaSOT and LaSOText; and AO on GOT-10k. Qualitative results in Fig. 3 show EdgeDAM under mobile deployment and benchmark evaluation scenarios.

4.3 Results on the DiDi Benchmark

DiDi [18] evaluates tracking under frequent semantic distractors. It contains 180 sequences sampled from GOT-10k [11], LaSOT [8], UTB180 [1], and VOT2020/2022 [13, 14], with more than one-third of the frames containing strong distractor overlap. This setting directly tests whether a tracker can preserve target identity when multiple visually similar candidates compete with the target.

Table 1: Comparison on the DiDi benchmark using three standard tracking metrics: Quality (overall tracking performance), IoU (measuring spatial accuracy), and Robustness (proportion of successfully tracked frames; higher is better).

Method	Quality	IoU	Robust.
SAMURAI	0.680	0.722	0.930
SAM2.1Long	0.646	0.719	0.883
ODTrack	0.608	0.740	0.809
Cutie	0.575	0.704	0.776
AOT	0.541	0.622	0.852
AQATrack	0.535	0.693	0.753
SeqTrack	0.529	0.714	0.718
KeepTrack	0.502	0.646	0.748
TransT	0.465	0.669	0.678
SAM2.1	0.649	0.720	0.887
SAM2.1++	0.694	0.727	0.944
EdgeDAM (Ours)	**0.741**	**0.730**	**0.973**

Table 2: Comparison of EdgeDAM with state-of-the-art trackers on the VOT2022 and VOT2020 benchmarks.

(a) VOT2022 benchmark results

Method	EAO	IoU	Robustness
MSAOT	0.673	0.781	0.956
DiffusionTrack	0.634	-	-
DAMTMask	0.624	0.796	0.891
MixFormerM	0.589	0.799	0.878
OSTrackSTS	0.581	0.775	0.867
Linker	0.559	0.772	0.861
SRATransTS	0.547	0.743	0.866
TransT_M	0.542	0.743	0.865
GDFormer	0.538	0.744	0.861
TransLL	0.530	0.735	0.861
LWL_B2S	0.516	0.736	0.831
D3Sv2	0.497	0.713	0.827
SAM2.1	0.692	0.779	0.946
SAM2.1++	0.753	0.800	0.969
EdgeDAM (Ours)	**0.790**	**0.811**	**0.974**

(b) VOT2020 benchmark results

Method	EAO	IoU	Robustness
ODTrack	0.605	0.761	0.902
MixViT-L+AR	0.584	0.755	0.890
SeqTrack-L	0.561	-	-
MixFormer-L	0.555	0.762	0.855
RPT	0.530	0.700	0.869
OceanPlus	0.491	0.685	0.842
AlphaRef	0.482	0.754	0.777
AFOD	0.472	0.713	0.795
SAM2.1	0.681	0.778	0.941
SAM2.1++	0.729	0.799	0.961
EdgeDAM (Ours)	**0.764**	**0.849**	**0.965**

Table 1 reports Quality, IoU, and Robustness. EdgeDAM achieves the highest Quality (0.741) and Robustness (0.973), while maintaining competitive IoU (0.730). Compared with SAM2.1++, EdgeDAM improves Quality by +0.047 and Robustness by +0.029, with a marginal IoU gain of +0.003. Compared with ODTrack, which obtains the highest IoU among the listed baselines (0.740), EdgeDAM trades only -0.010 IoU for substantially higher Quality (+0.133) and Robustness (+0.164). These results indicate that EdgeDAM primarily improves tracking continuity and target recovery under distractor ambiguity, rather than optimizing only per-frame overlap.

Unlike segmentation-memory trackers that rely on dense mask propagation or attention-based memory retrieval, EdgeDAM achieves this robustness through compact box-level target distractor memory. RAM preserves verified target evidence, while DRM penalizes hard negative candidates during re-identification.

4.4 Results on VOT Benchmarks

We evaluate EdgeDAM on VOT2020 [13] and VOT2022 [14]. Both benchmarks report Expected Average Overlap (EAO), Accuracy/IoU, and Robustness, jointly measuring spatial

Table 3: Results on three bounding-box benchmarks. AUC is reported for LaSOT and LaSOText, and AO is reported for GOT-10k.

Method	LaSOT	LaSOText	GOT10k
MixViT	0.724	-	0.757
LORAT	0.751	0.566	0.782
ODTrack	0.740	0.539	0.782
DiffusionTrack	0.723	-	0.747
DropTrack	0.718	0.527	0.759
SeqTrack	0.725	0.507	0.748
MixFormer	0.701	-	0.712
GRM-256	0.699	-	0.734
ROMTrack	0.714	0.513	0.742
OSTrack	0.711	0.505	0.737
KeepTrack	0.671	0.482	-
TOMP	0.685	-	-
SAM2.1	0.700	0.569	0.807
SAM2.1++	0.751	0.609	0.811
EdgeDAM (Ours)	**0.775**	**0.611**	**0.826**

precision and tracking continuity under failure-prone target states.

On VOT2020, EdgeDAM obtains an EAO of 0.764, surpassing SAM2.1++ (0.729) and SAM2.1 (0.681). EdgeDAM also improves IoU from 0.799 to 0.849 over SAM2.1++ and increases Robustness from 0.961 to 0.965. The larger gain in IoU shows that box-level state verification improves localization accuracy, while the robustness gain indicates fewer drift-induced failures.

On VOT2022, EdgeDAM achieves an EAO of 0.790, improving over SAM2.1++ (0.753) and the listed VOT2022 baselines in Table 2. It also reaches 0.811 IoU and 0.974 Robustness, compared with 0.800 IoU and 0.969 Robustness for SAM2.1++. The consistent improvement across EAO, IoU, and Robustness suggests that the proposed target–distractor memory factorization improves both candidate selection and recovery stability.

4.5 Results on Bounding-Box Benchmarks

We evaluate EdgeDAM on LaSOT [8], LaSOText [9], and GOT-10k [11]. We report AUC for LaSOT/LaSOText and AO for GOT-10k.

On LaSOT, EdgeDAM achieves 0.775 AUC, outperforming SAM2.1++ and LORAT, both at 0.751. The gain indicates that reliability-constrained RAM updates reduce long-term drift under scale variation, occlusion, and appearance change. On LaSOText, EdgeDAM obtains 0.611 AUC, slightly above SAM2.1++ (0.609) and higher than LORAT (0.566) and ODTrack (0.539). Since LaSOText evaluates unseen categories, this result supports the category-agnostic design of EdgeDAM, where class-collapsed candidate generation is combined with box-level target recovery.

On GOT-10k, EdgeDAM achieves 0.826 AO, improving over SAM2.1++ (0.811), LORAT (0.782), and ODTrack (0.782). Under the strict category-level separation protocol, EdgeDAM maintains strong generalization by relying on reliability-verified target states and distractor suppression rather than category-specific semantics.

4.6 Ablation Study

We analyze EdgeDAM along three axes: component contribution, DAM capacity, and multi-distractor recovery. Unless otherwise stated, ablations are conducted on a 35K-frame horse-rider dataset with frequent partial and full occlusions. We also perturb all thresholds and

Table 4: Component ablation on the custom dataset.

Variant	Det.	RAM	DRM	Held	IoU	Rec.	FPS
CSRT only					0.612	0.18	49.3
+Detector	✓				0.781	0.53	30.4
+RAM	✓	✓			0.874	0.71	28.6
+DRM	✓	✓	✓		0.912	0.89	26.9
Full EdgeDAM	✓	✓	✓	✓	**0.930**	**0.97**	**25.8**

Rec. denotes recovered occlusion fraction.

Table 5: DAM capacity ablation on the custom dataset.

RAM–DRM	IoU	AreaTol	RecF	FPS
5–5	0.910	1.15	2	24.89
10–10	**0.930**	**1.20**	**2**	**25.81**
15–15	0.870	1.25	2	19.19
20–20	0.880	1.30	2	16.38

RecF denotes mean recovery latency in frames.

Table 6: Multi-occlusion recovery under increasing distractor density. DAM denotes RAM–DRM capacity.

Scene	DAM	Occlusions	Recoveries
1 distractor	5–5	1	1
1 distractor	10–10	2	2
1 distractor	15–15	3	3
2 distractors	5–5	1	1
2 distractors	10–10	2	2
2 distractors	15–15	3	3
3 distractors	5–5	1	1
3 distractors	10–10	2	2
3 distractors	15–15	3	2

scoring weights by ±20% on DiDi; the resulting IoU variation remains below 0.015, indicating limited sensitivity to moderate hyperparameter changes.

4.7 Component Analysis

Table 4 isolates the contribution of each module. CSRT alone fails under occlusion, yielding low IoU and limited recovery. Adding the detector improves re-alignment but remains vulnerable to distractor switches. RAM improves stability by admitting only verified target states, while DRM further improves recovery by suppressing hard negative candidates. The full EdgeDAM configuration achieves the best IoU, recovery rate, and real-time throughput, showing that RAM, DRM, and held-box stabilization provide complementary gains.

4.8 DAM Capacity

Table 5 evaluates RAM–DRM capacity. The 10–10 setting gives the best IoU/FPS trade-off and is used by default. Larger buffers reduce accuracy and throughput: stale or partially corrupted entries remain longer under FIFO eviction and compete with valid anchors during recovery. Thus, increasing memory capacity does not monotonically improve box-level re-identification.

4.9 Controlled multi-distractor stress test

Table 6 evaluates recovery under increasing distractor density. EdgeDAM recovers all occlusions in one- and two-distractor scenes. With three distractors, recovery remains stable for 5–5 and 10–10 memories, while the 15–15 setting misses one recovery due to stale anchor competition. This result is consistent with Table 5: moderate memory capacity improves recovery, whereas oversized buffers increase distractor ambiguity.

5 Conclusion

We presented EdgeDAM, a real-time tracking framework that reformulates distractor-aware memory for bounding-box tracking. Instead of maintaining dense masks or memory tokens, EdgeDAM factorizes memory into reliability-verified target states and hard negative distractor evidence, enabling efficient re-identification under occlusion and distractor interference. Confidence-driven switching and held-box stabilization further protect memory from uncertain observations, improving recovery without dense feature retrieval or transformer memory attention. Experiments across six benchmarks show that box-level target–distractor memory provides a practical robustness–efficiency trade-off for mobile tracking. Future work will extend this formulation to multi-object and 3D occlusion-aware tracking.

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